

AZ-104

Tag 3

Administer Azure Storage

Guten Morgen!



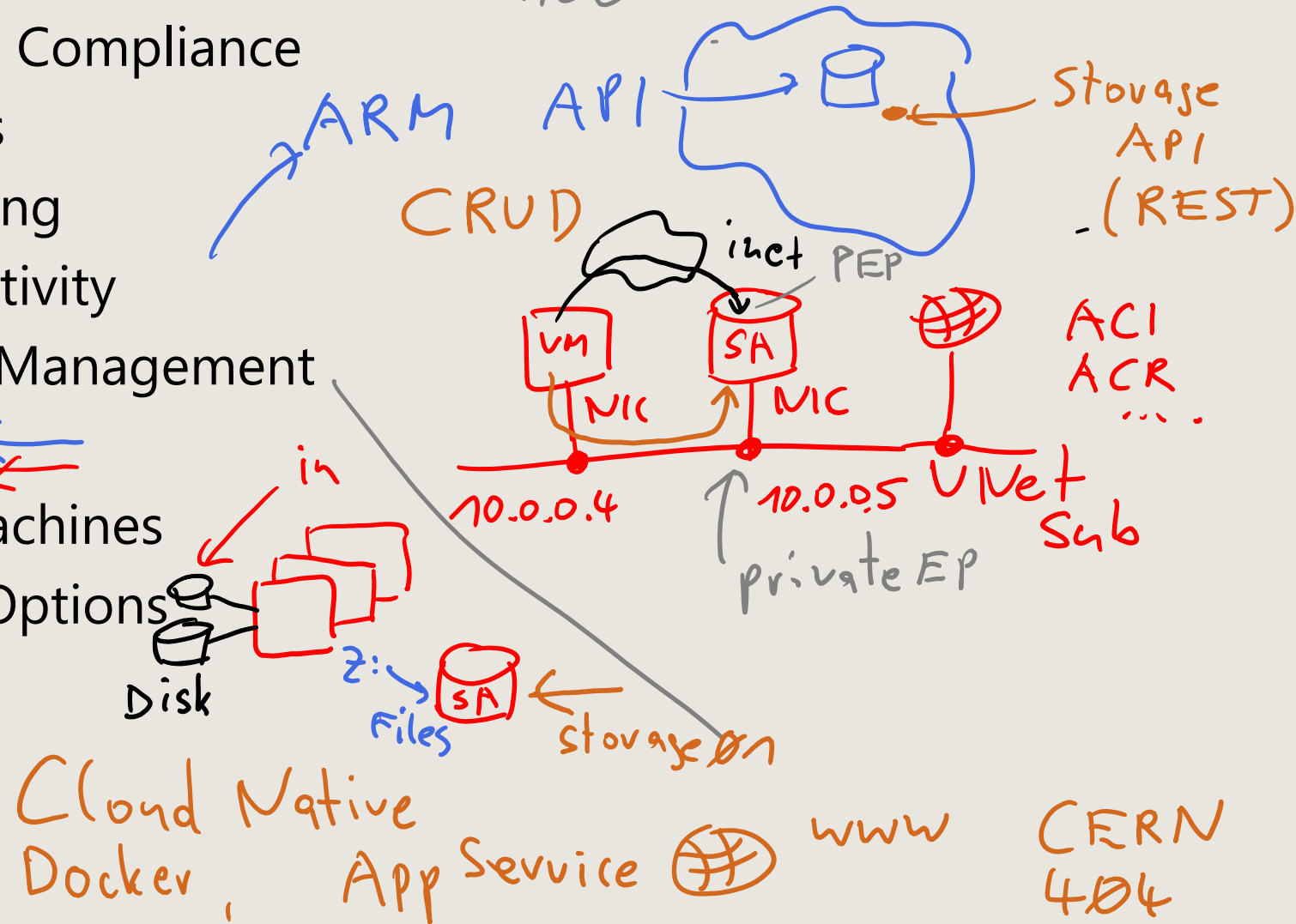
AZ-104 Agenda

- 01: Administer Identity
- 02: Administer Governance and Compliance
- 03: Administer Azure Resources
- 04: Administer Virtual Networking
- 05: Administer Intersite Connectivity
- 06: Administer Network Traffic Management
- 07: Administer Azure Storage
- 08: Administer Azure Virtual Machines
- 09: Administer PaaS Compute Options
- 10: Administer Data Protection
- 11: Administer Monitoring

△ Entra ID

Microsoft Learn
SC-900
- 400
- 300 ← OAuth 2.0
- 100

12³⁰ - 13⁰⁰



Learning Objectives - Administer Azure Storage

- Configure Storage Accounts
- Configure Blob Storage
- Configure Storage Security
- Configure Azure Files
- Lab 07 – Manage Azure Storage

Configure Storage Accounts



Learning Objective - Configure Storage Accounts

- Explore Azure Storage Services
- Determine Storage Account Kinds
- Determine Replication Strategies
- Access Storage
- Secure Storage Endpoints
- Demonstration – Configure a storage account
- Learning Recap

Exam objectives

Configure and manage storage accounts

- Create and configure storage accounts
- Configure Azure Storage redundancy

Configure access to storage

- Configure Azure Storage firewalls and virtual networks

Explore Azure Storage Services

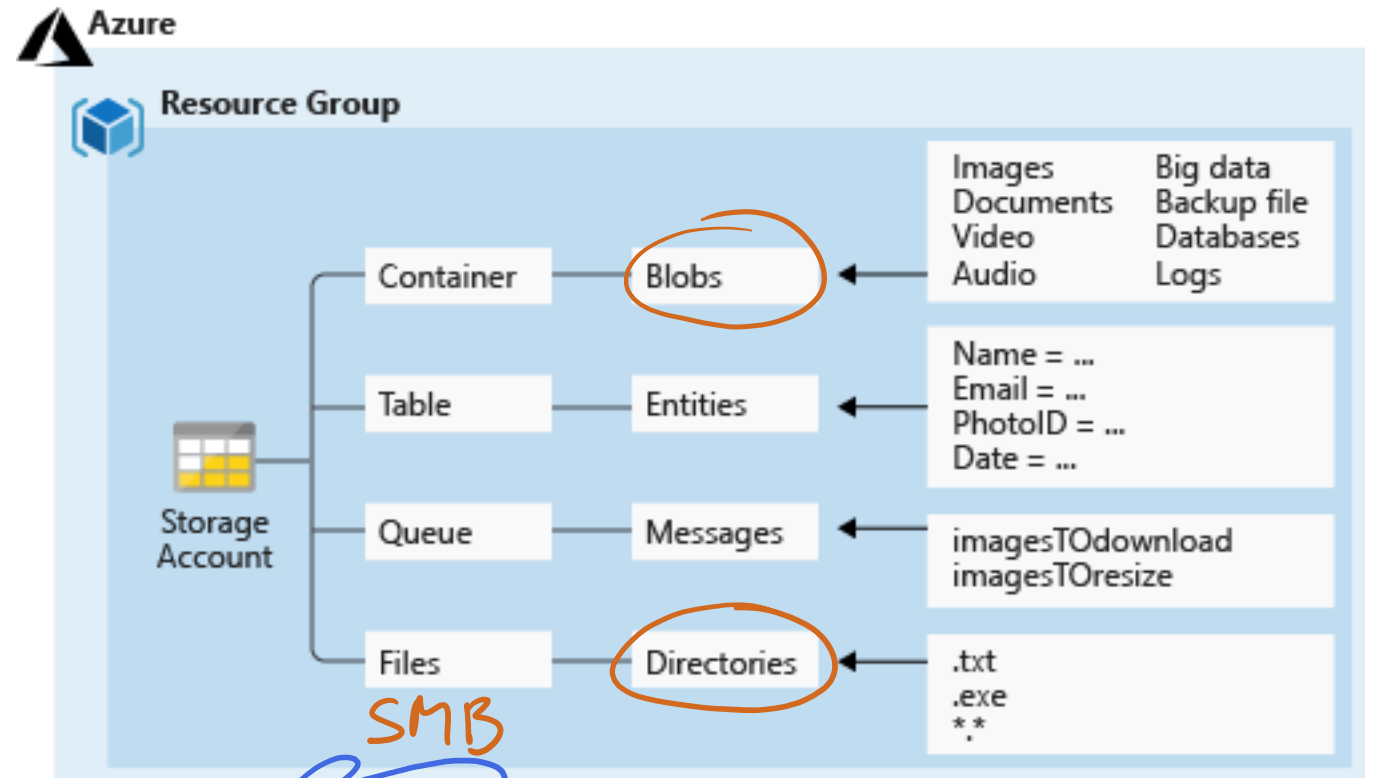
A service that you can use to store files, messages, tables, and other types of information

Azure Containers: A massively scalable object store for text and binary data

Azure Tables: Ideal for storing structured, non-relational data

Azure Queues: A messaging store for reliable messaging between application components

Azure Files: Managed file shares for cloud or on-premises deployments



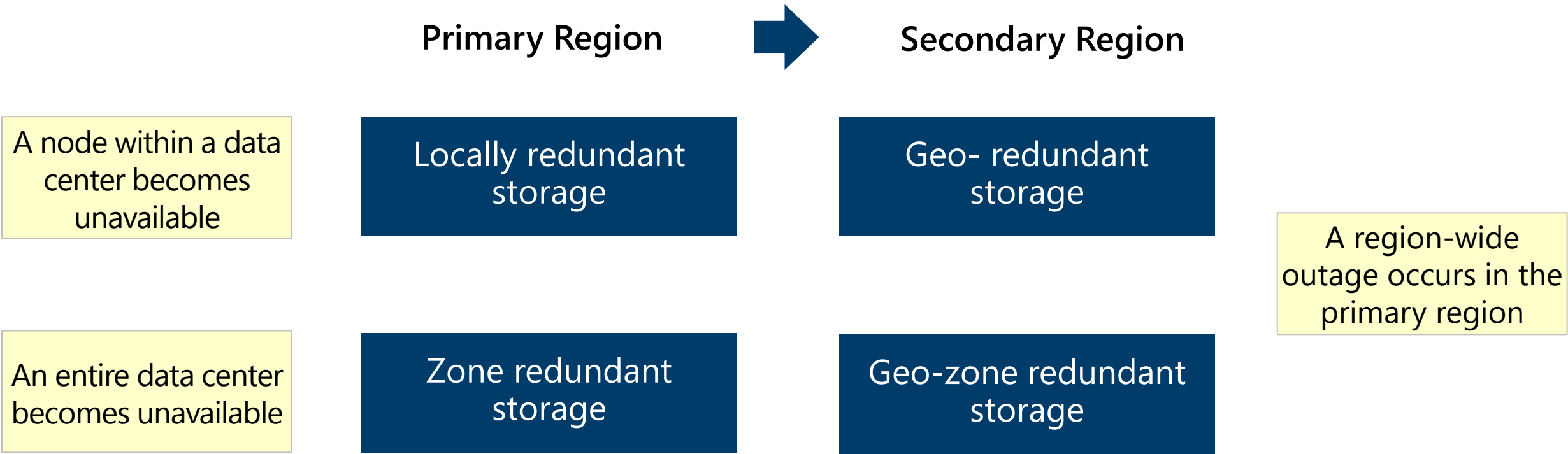
SMB
net use A: \\...

Determine Storage Account Kinds

All storage accounts are encrypted using Storage Service Encryption (SSE) for data at rest

Storage Account	Recommended usage
Standard general-purpose v2	Most scenarios including Blob, File, Queue, Table, and Data Lake Storage.
Premium block blobs	Block blob scenarios with high transactions rates, or scenarios that use smaller objects or require consistently low storage latency.
Premium file shares	Enterprise or high-performance file share applications.
Premium page blobs	Premium high-performance page blob scenarios.

Determine Replication Strategies



Access Storage

Every object has a unique URL address – based on account name and storage type

Container service: `https://mystorageaccount.blob.core.windows.net`

Table service: `https://mystorageaccount.table.core.windows.net`

Queue service: `https://mystorageaccount.queue.core.windows.net`

File service: `https://mystorageaccount.file.core.windows.net`

- If you prefer you can configure a custom domain name

CNAME record	Target
blobs.contoso.com	contosoblobs.blob.core.windows.net

Secure Storage Endpoints

Firewalls and Virtual Networks restrict access to the Storage Account from specific Subnets on Virtual Networks or public IP's

Subnets and Virtual Networks must exist in the same Azure Region or Region Pair as the Storage Account

Firewalls and virtual networks

Custom domain



Save



Discard



Refresh

Public network access



Enabled from all networks



Enabled from selected virtual networks and IP addresses



Disabled



All networks, including the internet, can access this storage account. [Learn more](#)

Network Routing

Determine how you would like to route your traffic as it travels from its source to an Azure endpoint. Microsoft routing is recommended for most customers.

Routing preference ⓘ



Microsoft network routing



Internet routing



The current combination of storage account kind, performance, replication, and location does not support network routing.

Configure Blob Storage



Learning Objective - Blob Storage

- Implement Blob Storage
- Create Blob Containers
- Create Blob Access Tiers
- Add Blob Lifecycle Management Rules
- Determine Blob Object Replication
- Demonstration – Configure Blob Storage
- Learning Recap

Exam objectives

Configure Azure Files and Azure Blob Storage

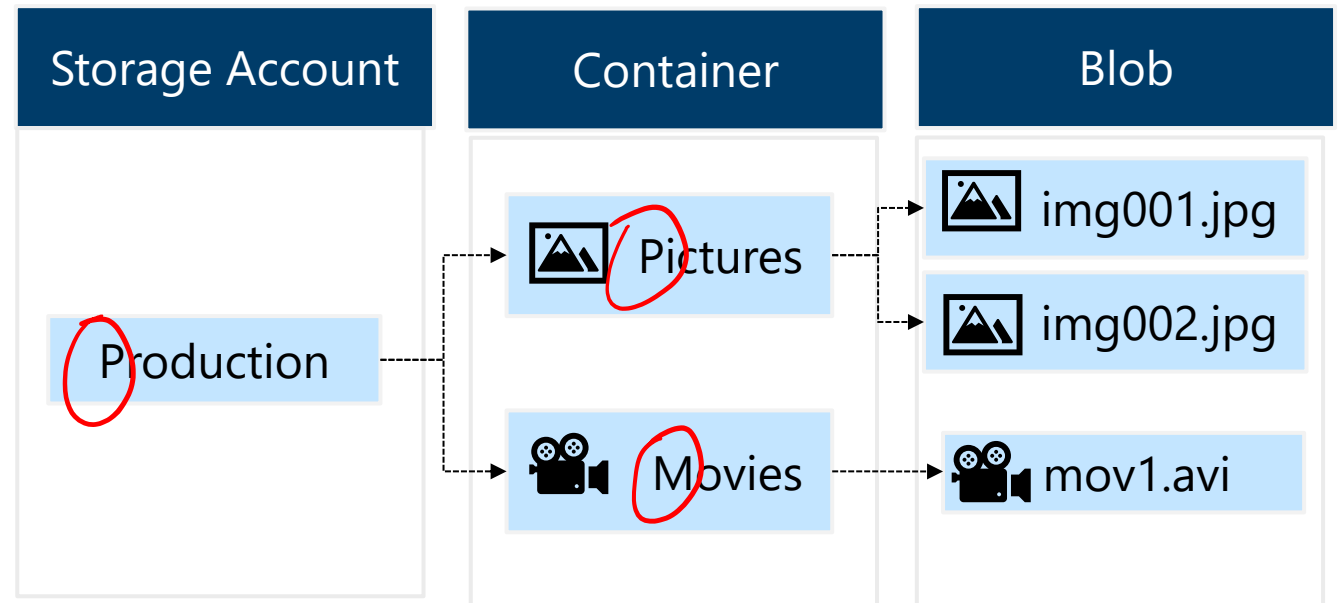
- Create and configure a container in Blob storage
- Configure storage tiers
- Configure blob lifecycle management
- Configure blob versioning

Implement Blob Storage

- Stores unstructured data in the cloud
- Can store any type of text or binary data
- Also referred to as *object storage*

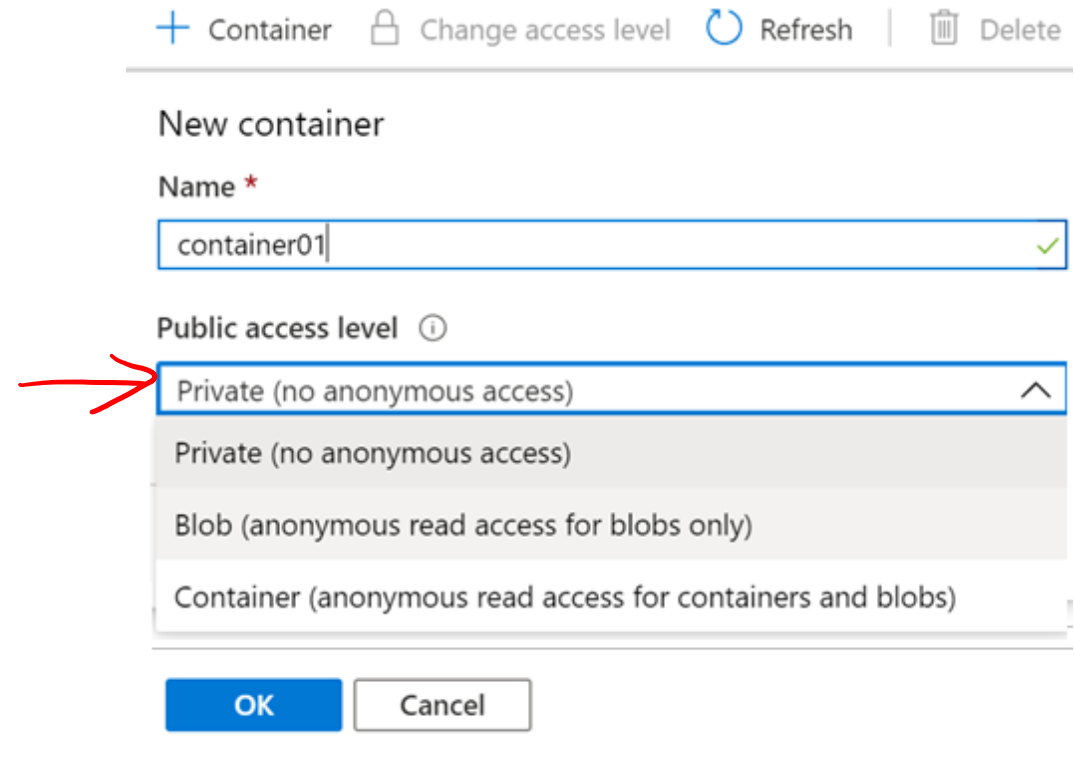
Common uses:

- Serving images or documents directly to a browser
- Storing files for distributed access
- Streaming video and audio
- Storing data for backup and restore, disaster recovery, archiving
- Storing data for analysis by an on-premises or Azure-hosted service



Create Blob Containers

- All blobs must be in a container
- Accounts have unlimited containers
- Containers can have unlimited blobs
- Restrict access using the public access level



Container management toolbar: + Container, Change access level, Refresh, Delete

New container

Name *

container01 ✓

Public access level ⓘ

Private (no anonymous access) ^

Private (no anonymous access)

Blob (anonymous read access for blobs only)

Container (anonymous read access for containers and blobs)

OK Cancel

Create Blob Access Tiers

Hot tier – Data that is accessed or modified frequently

Cool tier – Data that is infrequently accessed or modified and stored for at least 30 days

Cold tier – Data that is infrequently accessed or modified and stored for at least 90 days

Archive – Data that can tolerate several hours of retrieval latency and will remain in the Archive tier for at least 180 days

12h

Change tier

infoicon.jpg



Optimize storage costs by placing your data in the appropriate access tier. [Learn more](#)

Access tier

Hot (Inferred)



Hot (Inferred)

Cool

Cold

Archive

Add Blob Lifecycle Management Rules

- Transition blobs to a cooler storage tier to optimize for performance and cost
- Delete blobs at the end of their lifecycle
- Apply rules for lifecycle management tasks

Add a rule ...

✓ Details 2 Base blobs 3 Filter set

Lifecycle management uses your rules to automatically move blobs to cooler tiers or to delete them. If you create multiple rules, the associated actions must be implemented in tier order (from hot to cool storage, then archive, then deletion).

If

Base blobs were *

☒ Last modified

☐ Created

More than (days ago) *

Enter a value

Then

Delete the blob

Move to cool storage

For infrequently accessed data that you want to keep on cool storage for at least 30 days.

Move to cold storage

For rarely accessed data that you want to keep for at least 90 days.

Move to archive storage

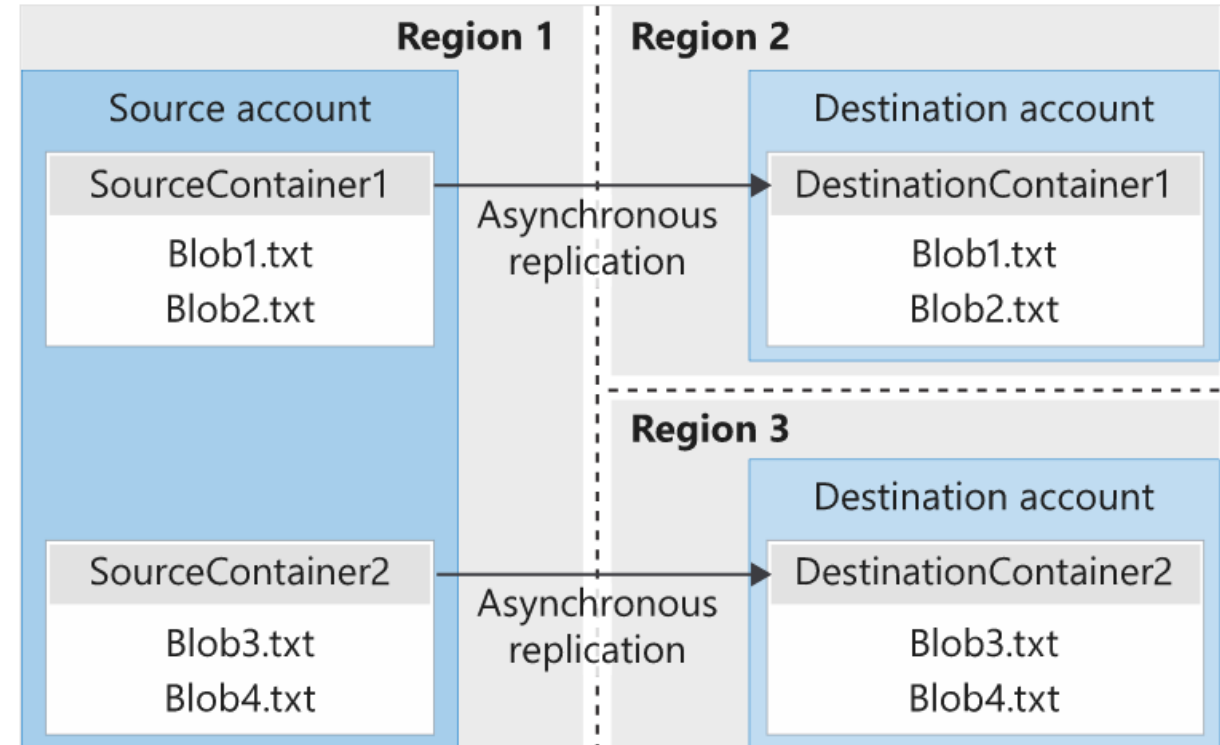
Use if you don't need online access and want to keep the object for 180 days or longer.

Delete the blob

Deletes the object per the specified conditions.

Determine Blob Object Replication

- Asynchronous to any other Region
- Minimizes latency for read requests
- Increases efficiency for compute workloads
- Optimizes data distribution
- Optimizes costs



Configure Storage Security



Learning Objective - Configure Storage Security

- Review Storage Security Strategies
- Create Shared Access Signatures
- Identify URI and SAS Parameters
- Demonstration – Configure Storage Security
- Determine Storage Service Encryption
- Create Customer Managed Keys
- Apply Storage Security Best Practices
- Learning Recap

Exam objectives

Configure access to storage

- Create and use shared access signature (SAS) tokens
- Configure stored access policies
- Manage access keys

Configure and manage storage accounts

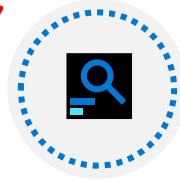
- Configure storage account encryption

Review Storage Security Strategies

in Rest — Microsoft
Bring your own key

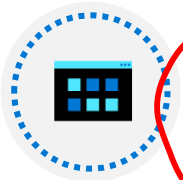


Storage Service Encryption



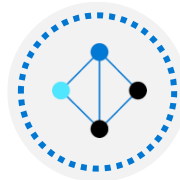
Shared Access Signatures –
delegated access

SAS



Authentication with Entra ID and
RBAC

App Registr.

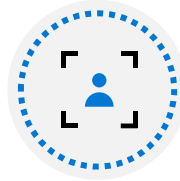


Shared Key – encrypted
signature string

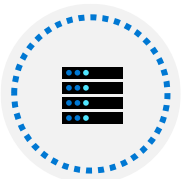
*Key 1
2*



Client-side encryption, HTTPS,
and SMB 3.0 for data in transit



?
Anonymous access to containers
and blobs



Azure disk encryption

Create Shared Access Signatures

- Provides delegated access to resources
- Grants access to clients without sharing your storage account keys
- The account SAS delegates access to resources in one or more of the storage services
- The service SAS delegates access to a resource in just one of the storage services

Signing method ⓘ
☒ Account key ☐ User delegation key

Signing key ⓘ
Key 1 ▼

Permissions * ⓘ
Read ▼

Start and expiry date/time ⓘ
Start
02/01/2021 
(UTC-08:00) Coordinated Universal Time-08 ▼

Expiry
02/02/2021 
(UTC-08:00) Coordinated Universal Time-08 ▼

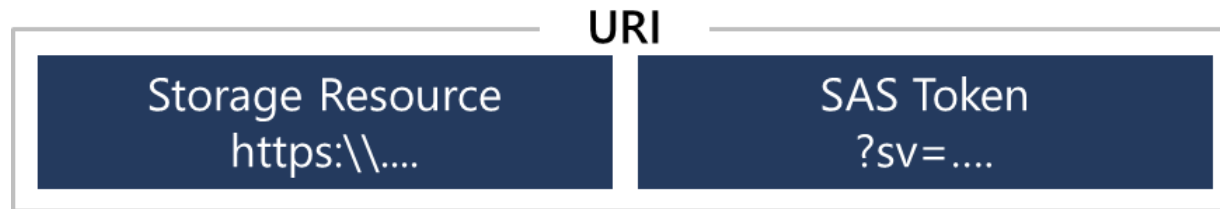
Allowed IP addresses ⓘ
for example, 168.1.5.65 or 168.1.5.65-168.1....

Allowed protocols ⓘ
☒ HTTPS ☐ HTTP

Generate SAS token and URL

Identify URI and SAS Parameters

- A SAS is a signed URI that points to one or more storage resources
- Consists of a storage resource URI and the SAS token



<https://myaccount.blob.core.windows.net/?sp=r&st=2020-05-11T18:31:43Z&se=2020-05-12T02:31:43Z&spr=https&sv=2019-10-10&sr=b&sig=j0qABJZHfUVEBQ3yVn7kWiCKl00sxCiK1rzEchfAz8U%3D>

Includes parameters for resource URI, storage services version, services, resource types, start time, expiry time, resource, permissions, IP range, protocol, signature

Demonstration – Configure storage security



- Generate a shared access signature
- Review the configuration permissions
- Use the SAS URL to test the permissions

Determine Storage Service Encryption

You can use your own key (next topic)

- Protects your data for security and compliance
- Automatically encrypts and decrypts your data
- Encrypted through 256-bit AES encryption
- Is enabled for all new and existing storage accounts and cannot be disabled
- Is transparent to users

Encryption

 Save  Discard

Storage service encryption protects your data at rest. Azure Storage encrypts your data as it's written in our datacenters, and automatically decrypts it for you as you access it.

By default, data in the storage account is encrypted using Microsoft Managed Keys. You may choose to bring your own key.

Please note that after enabling Storage Service Encryption, only new data will be encrypted, and any existing files in this storage account will retroactively get encrypted by a background encryption process.

[Learn More about Azure Storage Encryption](#) 

Encryption type

- ☒ Microsoft Managed Keys
- ☐ Customer Managed Keys

Create Customer Managed Keys

- Use the Azure Key Vault to manage your encryption keys
- Create your own encryption keys and store them in a key vault
- Use Azure Key Vault's APIs to generate encryption keys
- Custom keys give you more flexibility and control

Encryption type

☐ Microsoft Managed Keys

☒ Customer Managed Keys 

i The storage account named 'storage987123' will be granted access to the selected key vault. Both soft delete and purge protection will be enabled on the key vault and cannot be disabled. [Learn more about customer managed keys](#) 

Encryption key

☐ Enter key URI

☒ Select from Key vault

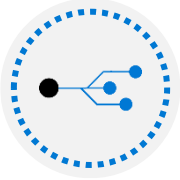
Key vault and key *

Key vault: keyvault987123

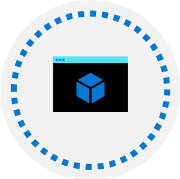
Key: storagekey

[Select a key vault and key](#)

Apply Storage Security Best Practices



Always use HTTPS to create or distribute a SAS



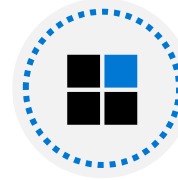
Reference stored access policies where possible



Use near-term expiration times on an ad hoc SAS



Use Storage Analytics to monitor your application



Be specific with the resource to be accessed



Understand that your account will be billed for any usage



Validate data written using SAS



Don't assume SAS is always the correct choice

Configure Azure Files

RD SMB 3

Learning Objective - Configure Azure Files

- Compare Storage for File Shares and Blob Data
- Manage File Shares
- Create File Share Snapshots
- Implement Soft Delete for Azure Files
- Use Azure Storage Explorer
- Demonstration – Configure File Shares
- Learning Recap

Exam objectives

Configure Azure Files and Azure Blob Storage

- Create and configure a file share in Azure Storage
- Configure snapshots and soft delete for Azure Files
- Manage data by using Azure Storage Explorer

Compare Storage for File Shares and Blob Data

Feature	Description	When to use
Azure Files	SMB interface, client libraries, and a REST interface that allows access from anywhere to stored files	• Lift and shift an application to the cloud • Store shared data across multiple virtual machines • Store development and debugging tools that need to be accessed from many virtual machines
Azure Blobs	Client libraries and a REST interface that allows unstructured data (flat namespace) to be stored and accessed at a massive scale in block blobs	• Support streaming and random-access scenarios • Access application data from anywhere

Manage File Shares

- Windows – ensure port 445 is open
- Linux – mount the drive
- MacOS – mount the drive
- Secure transfer required – SMB 3.0 encryption

Connect

publicwebsite



Windows

Linux

macOS

To connect to this Azure file share from Windows, choose from the following authentication methods and run the PowerShell commands from a normal (not elevated) PowerShell terminal:

Drive letter

Z



Authentication method



Active Directory or Microsoft Entra



Storage account key



Connecting to a share using the storage account key is only appropriate for admin access. Mounting the Azure file share with the Active Directory or Microsoft Entra identity of the user is preferred. [Learn more](#)

Show Script

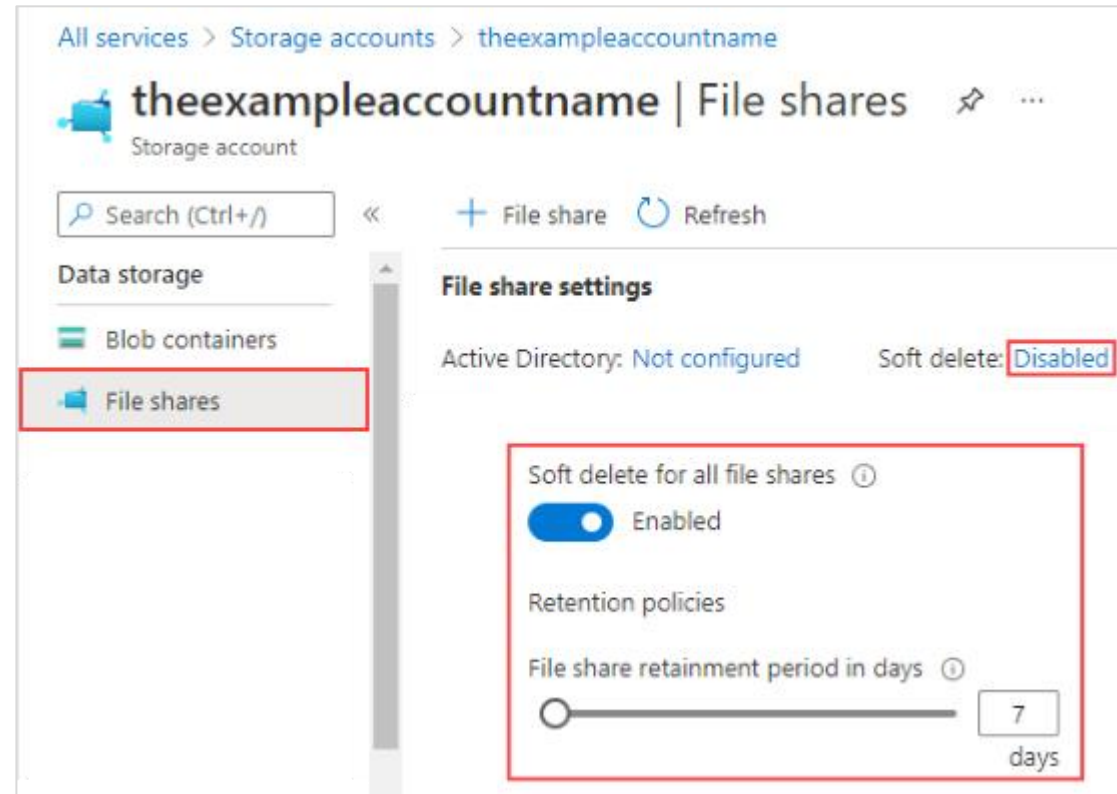
Create File Share Snapshots

- Protection against application error and data corruption
- Protection against accidental deletions or unintended changes
- Support backup and recovery
- Incremental snapshot that captures the share state at a point in time
- Snapshot at the *file share level*, and restore at the *file level*
- Is read-only copy of your data

 Add snapshot  Refresh  Delete		
Name	Date created	Initiator
☐ 2020-03-12T00:58:38.0000000Z	3/11/2020, 8:58:38 PM	-

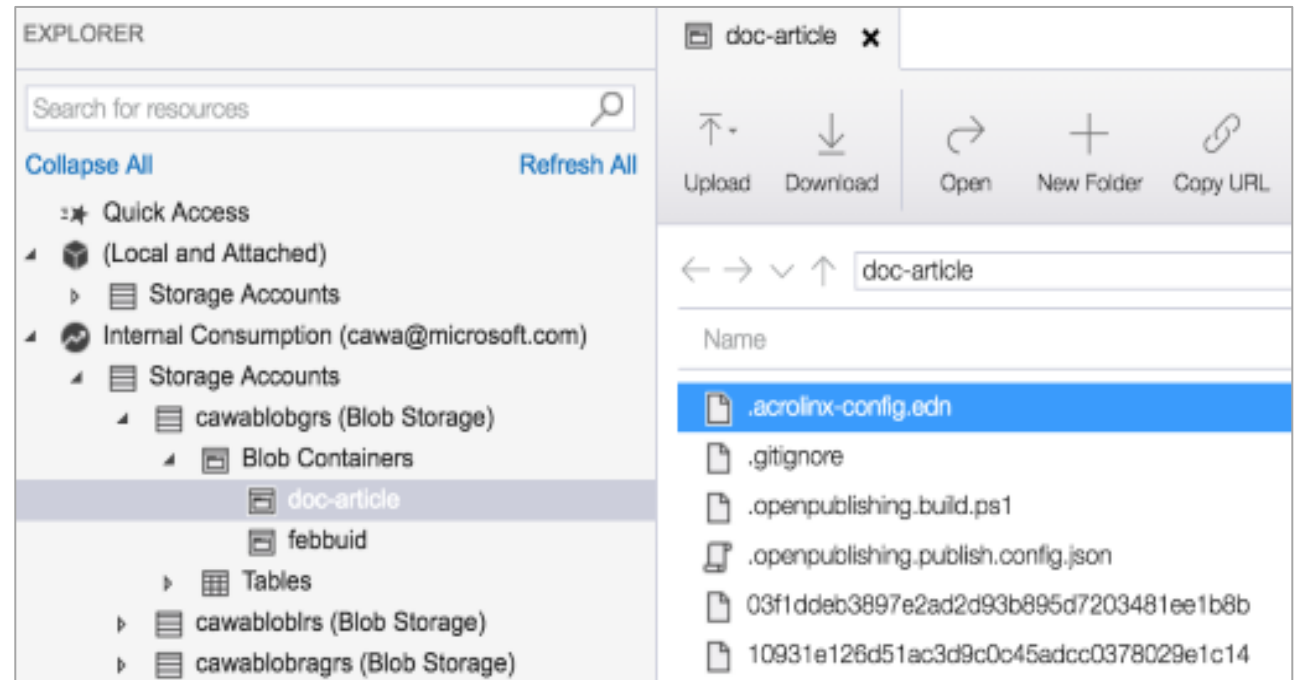
Implement Soft Delete for Azure Files

- Recovery from accidental data loss
 - Major change or upgrade scenarios
 - Business continuity – ransomware situations
 - Data compliance retention
-
- Enabled at the storage account level
 - Transitions content to a soft deleted state
 - Provides a retention period of 1 and 365 days
 - Works on new or existing file shares
 - Doesn't work for NFS shares



Use Azure Storage Explorer

- Download and install
- Access multiple accounts and subscriptions
- Create, delete, view, edit storage resources
- View and edit Blob, Queue, Table, File, Cosmos DB storage and Data Lake Storage
- Obtain shared access signature (SAS) keys
- Available for Windows, Mac, and Linux

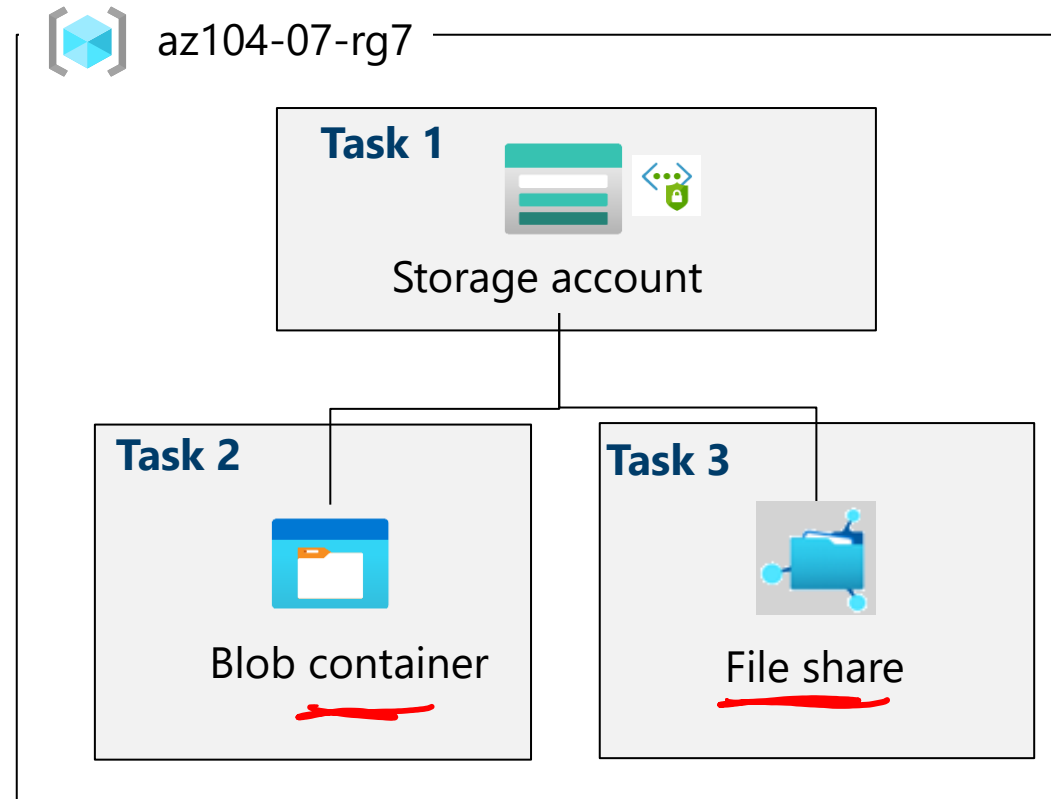


Also consider portal-based Azure Storage Browser and Azure Storage Mover

Lab – Manage Azure Storage



Lab 07 – Architecture diagram



End of presentation



